

Covariant and contravariant vectors

Chris Wang

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I keep getting contravariant and covariant tensors and vectors confused, so I decided to derive their transformation laws to remind myself. Typically, a *contravariant vector* is a thing that most people simply refer to as a vector, living in a vector space. Let's denote that $\mathbf{v}$. One might also call it a rank-1 tensor, a $(1, 0)$ -tensor, whatever. And it supposedly "transforms like a tensor". Meanwhile, a *covariant vector* is formally a linear form: it takes as input a regular old vector and spits out a scalar (real or complex number). Let's denote it ϕ , though, in the spirit of representing this linear form as a covariant vector, I'll also boldface it, so we'll name it $\boldsymbol{\phi}$. Some people also call this a rank-1 tensor, but now it's a $(0, 1)$ -tensor! Typically, the first entry in the 2-tuple specifying tensors tells you the contravariant index of the vector, while the second entry tells you the covariant index. Another way to put it is the first entry tells you how many covariant vectors it will take as input, and the second entry tells you how many contravariant vectors it will take as input.

Using more symbols to say the same thing, our vector $\mathbf{v}$ lives in a vector space V , while our covariant vector (also covector) $\boldsymbol{\phi}$ lives in something known as the dual space of V , V^* . That's not too relevant, though the relevant part is that $\boldsymbol{\phi}$ takes a vector and returns a scalar; in essence, we can write $\boldsymbol{\phi} : V \rightarrow \mathbb{F}$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. **We also typically represent contravariant vectors as column vectors and covariant vectors as row vectors.**

Anyways, covariant and contravariant tensors exist within a larger field of math known as differential geometry, and one of its special features is developing a coordinate-free representation of certain quantities. What that implies is that when you apply a change of basis, certain quantities don't change, such as a dot product. This can be useful for describing things in physics and shows up especially prominently in subjects like relativity, which is based on the principle that the laws of physics do not change under a change of reference frame. Since there are also many vector and dot-product like entities to consider in math, it's good to develop some intuition for how they transform under a change of basis.

Ok, enough rambling, let's start with regular old vectors. Suppose we apply a change of basis that takes us from the basis $\{\mathbf{e}_i\} \mapsto \{\mathbf{e}'_i\}$. Denote this change of basis matrix B , and if we represent the set of basis vectors as a block matrix with each basis vector as a column, we have that

$$\begin{bmatrix} \mathbf{e}'_1 & \mathbf{e}'_2 & \cdots & \mathbf{e}'_n \end{bmatrix} = \begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \cdots & \mathbf{e}_n \end{bmatrix} B. \quad (1)$$

Moreover, we can use this representation (and some slight abuse of notation) to write

$$\mathbf{v} = \begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_n \end{bmatrix} \cdot \mathbf{v} = \sum_i^n v_i \mathbf{e}_i. \quad (2)$$

Now, regardless of what coordinate system you use, a vector is still just a vector, and given two bases, it will

have two equivalent representations. In other words, in the transformed basis, we have

$$\mathbf{v} = \sum_i^n v'_i \mathbf{e}'_i = \begin{bmatrix} \mathbf{e}'_1 & \cdots & \mathbf{e}'_n \end{bmatrix} \cdot \mathbf{v}'. \quad (3)$$

We can now equate these two representations to show how a contravariant vector transforms under a change of basis.

$$\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_n \end{bmatrix} \cdot \mathbf{v} = \begin{bmatrix} \mathbf{e}'_1 & \cdots & \mathbf{e}'_n \end{bmatrix} \cdot \mathbf{v}' \quad (4)$$

$$= \begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_n \end{bmatrix} B \cdot \mathbf{v}' \quad (5)$$

$$\therefore \mathbf{v}' = B^{-1} \mathbf{v}. \quad (6)$$

Hence, a contravariant vector is one that transforms with the inverse of the change of basis matrix under a change of basis (or coordinate change). Let's look at covariant vectors now. Let the dot product between ϕ and $\mathbf{v}$ be $\phi(\mathbf{v}) = \phi \cdot \mathbf{v} = s$ for some $s \in \mathbb{F}$. This dot product is fixed under coordinate change, and thus we have $\phi \cdot \mathbf{v} = \phi' \cdot \mathbf{v}' = s$. We may also write

$$\phi = \phi \cdot \begin{bmatrix} \mathbf{e}^1 \\ \vdots \\ \mathbf{e}^n \end{bmatrix}, \quad (7)$$

and note that we use superscripts for the covariant tensor. This is just a particular convention. Just like above, we equate two representations under the change of basis.

$$\phi \cdot \mathbf{v} = \phi' \cdot \mathbf{v}' \quad (8)$$

$$= \phi' \begin{bmatrix} \mathbf{e}'^1 \\ \vdots \\ \mathbf{e}'^n \end{bmatrix} \cdot \begin{bmatrix} \mathbf{e}'_1 & \cdots & \mathbf{e}'_n \end{bmatrix} \mathbf{v}' \quad (9)$$

$$= \phi' \begin{bmatrix} \mathbf{e}'^1 \\ \vdots \\ \mathbf{e}'^n \end{bmatrix} \cdot \begin{bmatrix} \mathbf{e}'_1 & \cdots & \mathbf{e}'_n \end{bmatrix} B^{-1} \mathbf{v}. \quad (10)$$

Now, we assume that the bases we're using are orthonormal (and even if they're not, we can just apply QR decomposition to get an orthonormal basis), which means that the slightly cursed column-row product is actually just the n -dimensional identity matrix. Therefore, we have the equality

$$\phi \cdot \mathbf{v} = \phi' B^{-1} \mathbf{v} \implies \phi' = \phi B. \quad (11)$$

In other words, this means that a covariant vector transforms with the change of basis matrix under a change of basis.